1.4.10 d)

$$r_{yy}[k] + a_1 r_{yy}[k-1] + a_2 r_{yy}[k-2] = 0$$

$$r_{yy}[k] = -a_1 r_{yy}[k-1] - a_2 r_{yy}[k-2]$$

$$x[k] + a_1 x[k-1] + a_2 x[k-2] = 0$$

- $x[k] = C\gamma^k$
- $x[k-1] = c\gamma^{k-1}$
- $x[k-2] = c\gamma^{k-2}$

$$C\gamma^k + a_1 C\gamma^{k-1} + a_2 C\gamma^{k-2} = 0$$

$$(\gamma^2 + a_1\gamma + a_2)C\gamma^{k-2} = 0$$

$$\gamma^2 + a_1\gamma + a_2 = 0$$

$$x[k] = c_1\gamma_1^k + c_2\gamma_2^k$$

Initial condition:

- $r_{yy}[0] = \sigma_y^2$
- $r_{yy}[1] = -\dfrac{a_1}{1+a_2}\sigma^2$

$\vdots$

$$\ddot{x}(t) + a_1\dot{x}(t) + a_2 x(t) = 0$$

Assume: $x(t) = Ce^{\lambda t}$

- $\dot{x}(t) = \lambda c e^{\lambda t}$
- $\ddot{x}(t) = \lambda^2 c e^{\lambda t}$

$$(\lambda^2 + a_1\lambda + a_2)Ce^{\lambda t} = 0$$

$$(\lambda^2 + a_1\lambda + a_2) = 0$$

↳ roots: λ_1, λ_2

$$\begin{cases} x(t) = c_1 e^{\lambda_1 t} \\ \text{or} \\ x(t) = c_2 e^{\lambda_2 t} \end{cases}$$

$$x(t) = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t}$$

1.4-7

$$y[t] = f[t] + 2f[t-1] + 3f[t-2]$$

$$\begin{bmatrix} E[y[t]y[t]] & E[y[t]y[t-1]] & E[y[t]y[t-2]] \\ E[y[t-1]y[t]] & E[y[t-1]y^T[t-1]] & E[y[t-1]y[t-2]] \\ E[y[t-2]y[t]] & E[y[t-2]y[t-1]] & E[y[t-2]y[t-2]] \end{bmatrix}$$

$$y[t] = \frac{1}{3}\left(f[t] + f[t-1] + f[t-2]\right)$$

$$① = E[y[t]^2]$$

$$= E[(f[t] + 2f[t-1] + 3f[t-2])^2]$$

$$= E[f[t]^2 + 4f[t]f[t-1] + 6f[t]f[t-2] + 4f[t-1]^2 + 12f[t-1]f[t-2] + 9f[t-2]^2]$$

$$= 0.1 + (4)(0.1) + 9(0.1)$$

$$\sigma_f^2 = E[f[t]^2] - (E[f[t]])^2$$

$$= E[f[t]^2]$$

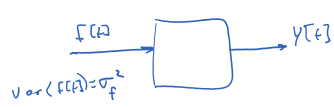
$$= 0.1$$

1.4-8

$$y[t+1] + a_1 y[t] = f[t+1]$$

$$y[t+1] = -a_1 y[t] + f[t+1]$$

AR(1)



$$Var(y) = E[y[t]^2] - \underbrace{(E[y[t]])}^2$$

$$y[t] = -a_1 y[t-1] + f[t]$$

$$E[y[t]] = E[-a_1 y[t-1] + f[t]]$$
$$= E[-a_1 y[t-1]] + E[f[t]]$$
$$= -a_1 E[y[t-1]] + 0$$
$$= -a_1 E[y[t]]$$
$$E[y[t]] = 0.$$

$$\sigma_y^2 = E[y[t]^2]$$
$$= E[(-a_1 y[t-1] + f[t])^2]$$
$$= E[a_1^2 y[t-1]^2 - 2a_1 y[t-1] f[t] + f[t]^2]$$
$$= a_1^2 E[y[t-1]^2] - 2a_1 \cdot 0 + \sigma_f^2$$
$$\sigma_y^2 = a_1^2 E[y[t-1]^2] + \sigma_f^2$$
$$\sigma_y^2 = a_1^2 \sigma_y^2 + \sigma_f^2$$

1.4.10

$$y[k+2] + a_1 y[t+1] + a_2 y[t] = f[t+2]$$

write as

$$y[t] = -a_1 y[t-1] - a_2 y[t-2] + f[t]$$

- mult. by $y^*[t-1]$: $E[y[t] y^*[t-1]] = E[-a_1 [y[t-1] y^*[t-1]] - a_2 [y[t-2] y^*[t-1]] + f[t] y^*[t-1]]$

Take E

$$* \quad r_{yy}[1] = -a_1 r_{yy}[0] - a_2 r_{yy}[-1]$$

- mult. by $y^*[t-2]$: $E[y[t] y^*[t-2]] = E[-a_1 y[t-1] y^*[t-2] - a_2 y[t-2] y^*[t-2] + f[t] y^*[t-2]]$
- Take E

$$* \quad r_{yy}[2] = -a_1 r_{yy}[1] - a_2 r_{yy}[0]$$

$$\begin{cases} r_{yy}[1] = -a_1 r_{yy}[0] - a_2 r_{yy}[-1] \\ r_{yy}[2] = -a_1 r_{yy}[1] - a_2 r_{yy}[0] \end{cases}$$

$$\begin{bmatrix} r_{yy}[0] & r_{yy}[-1] \\ r_{yy}[1] & r_{yy}[0] \end{bmatrix} \begin{bmatrix} -a_1 \\ -a_2 \end{bmatrix} = \begin{bmatrix} r_{yy}[1] \\ r_{yy}[2] \end{bmatrix}$$

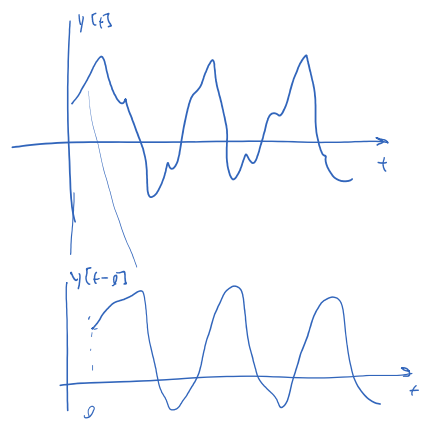
$$r_{yy}[-1] = r_{yy}[1]$$

$$r_{yy}[l-k] = E[y[t-k] y^*[t-l]]$$

Has no t

e.g. $k=0$:

$$r_{yy}[l] = E[y[t] y^*[t-l]]$$



$$\boxed{\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}}$$